

Rebuttal for CoRL Submission #18

We thank the reviewers and Area Chair for identifying important ambiguities in the optimization, masking, ablations, and real-robot protocol. We agree with these concerns and describe concrete corrections and controlled experiments below. The current results are preliminary sanity checks; final controlled results will replace the marked entries.

TABLE I

SWIPE success/rollouts with 95% Wilson intervals. “Pilot” denotes the current hand-stabilized sanity runs; the aligned fixed-mount evaluation is ongoing.

Policy	Pilot ($N=5$)	Fixed mount
Legacy VPCE-DP (reference)	4/5 [0.38, 0.96]	TBD
Clean full, $\alpha=2$	2/5 [0.12, 0.77]	TBD
Clean full, $\alpha=1$	3/5 [0.23, 0.88]	TBD
Clean no-gating	0/5 [0.00, 0.43]	TBD
Phase-conditioned DP	2/5 [0.12, 0.77]	TBD
Progress-conditioned DP	3/5 [0.23, 0.88]	TBD

The pilot is reported for transparency, not as a model ranking; the first clean-full batch used a lower execution speed, while the legacy reference used a different workspace clip. The final column uses one frozen controller, identical near-contact initialization, and a rigidly mounted reader.

Q1: Is VPCE only task progress or phase? We trained same-backbone Phase-DP and Progress-DP baselines that replace the VPCE horizon with a progress-derived four-bin phase or normalized trajectory progress and do not use tactile-derived contact supervision. Their pilot point estimates are comparable to the clean full models, and all intervals are wide; therefore these data do not yet establish that VPCE is superior to explicit progress. We will base the revision on the aligned fixed-mount results. If the

baselines remain competitive, we will narrow the claim: on SWIPE, contact timing overlaps task phase, while VPCE predicts it online from RGB and motion without requiring oracle trajectory progress.

Q2–Q3: Loss weighting and proprioceptive masking. We agree that a learned scalar optimized through the same weighted denoising objective can be driven to its lower bound. We replace it with the fixed, bounded schedule $w(c) = \text{clip}(1 + \alpha c, 1, w_{\max})$ and test $(\alpha, w_{\max}) = (2, 3)$ and $(1, 2)$. We also agree that the original post-encoder low-dimensional mask was too coarse. The revised policy always retains proprioception and gripper width and hard-masks only tactile/contact features. These are direct implementation corrections rather than a defense of the original choices.

Q4: Is feature gating necessary? The submitted pilot gave 9/10 for both full and no-gating variants, suggesting that gating may be auxiliary. The revised no-gating sanity run (0/5) points in the opposite direction, but its small sample and the pilot protocol do not support a causal conclusion. We therefore will not claim that gating is either necessary or dispensable until the aligned results are complete, and will move the component ablations into the main text.

Human-held props and limitations. We agree that hand compliance can assist alignment, even without deliberate trajectory correction. We therefore separate hand-stabilized pilots from the rigid-mount evaluation and use the latter to answer this concern. Every SWIPE trial uses a 60-s timeout and succeeds only when the reader’s internal roller completes one effective rotation. Finally, VPCE is target-adapted using task-specific tactile demonstrations and calibrated contact labels; we do not claim zero-shot transfer to unseen tasks, objects, or viewpoints.